

$\pi_{0.6}^*$: a VLA That Learns From Experience

Physical Intelligence

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<https://pi.website/blog/pistar06>

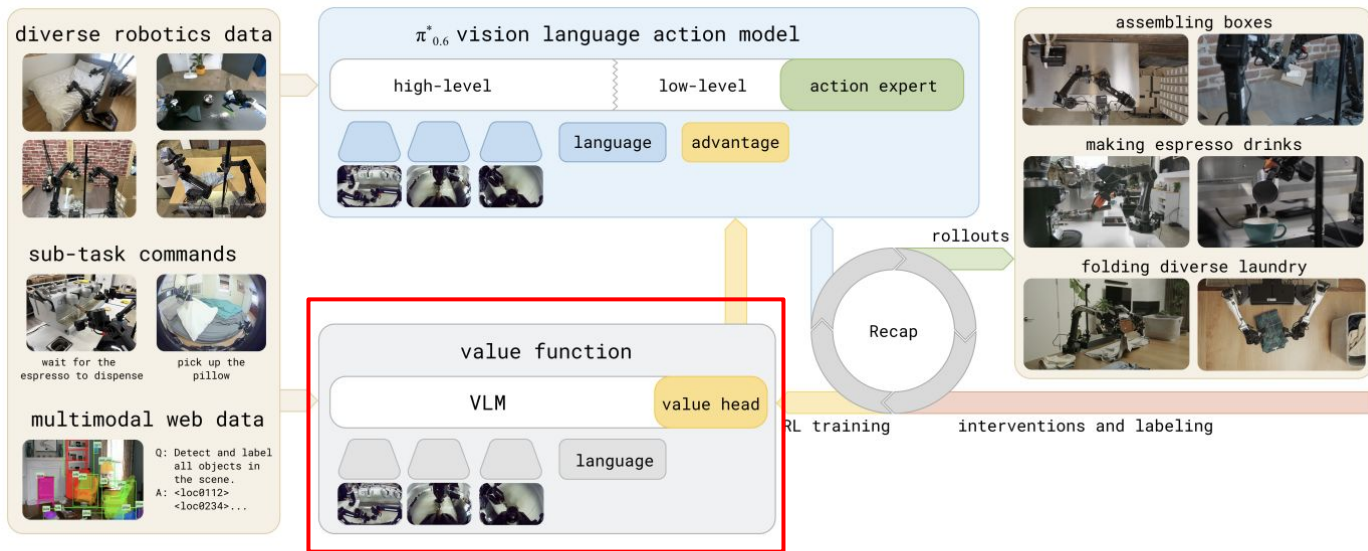


Fig. 1: **RECAP enables training VLAs with reward feedback and interventions.** Our system starts with a pre-trained VLA that incorporates *advantage conditioning*, allowing the model to learn effectively from real-world experience. For each task, we deploy the model and collect both autonomous rollouts and online human corrections. We then fine-tune the value function on this online data, improving its estimates of how actions influence performance. Fine-tuning and conditioning the VLA on these updated advantage estimates in turn improves policy behavior.

Reward-Conditioned Policies

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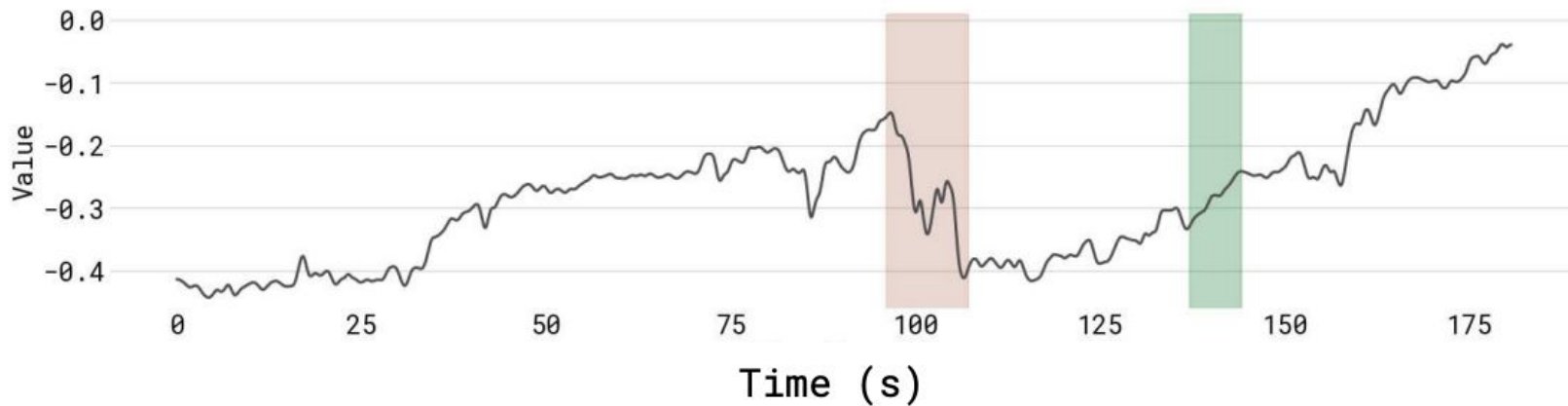
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Abstract

Reinforcement learning offers the promise of automating the acquisition of complex behavioral skills. However, compared to commonly used and well-understood supervised learning methods, reinforcement learning algorithms can be brittle, difficult to use and tune, and sensitive to seemingly innocuous implementation decisions. In contrast, imitation learning utilizes standard and well-understood supervised learning methods, but requires near-optimal expert data. Can we learn effective policies via supervised learning without demonstrations? The main idea that we explore in this work is that non-expert trajectories collected from sub-optimal policies can be viewed as optimal supervision, not for maximizing the reward, but for matching the reward of the given trajectory. By then conditioning the policy on the numerical value of the reward, we can obtain a policy that generalizes to larger returns. We show how such an approach can be derived as a principled method for policy search, discuss several variants, and compare the method experimentally to a variety of current reinforcement learning methods on standard benchmarks.

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Successful Episode: Folding Laundry



$$A^\pi(\mathbf{o}_t, \mathbf{a}_t) = \mathbb{E}_{\rho_\pi(\tau)} \left[\sum_{t'=t}^{t+N-1} r_{t'} + V^\pi(\mathbf{o}_{t+N}) \right] - V^\pi(\mathbf{o}_t),$$

*intuition: first derivative
of the value
conditioned on the
action*

A = 0: $\mathbf{a}_t$ is **as good as** the policy
(~same reward and ~same end state)

A < 0: $\mathbf{a}_t$ is **worse** than the policy
(< reward and/or < end state)

A > 0: $\mathbf{a}_t$ is **better** than the policy
(> reward and/or > end state)

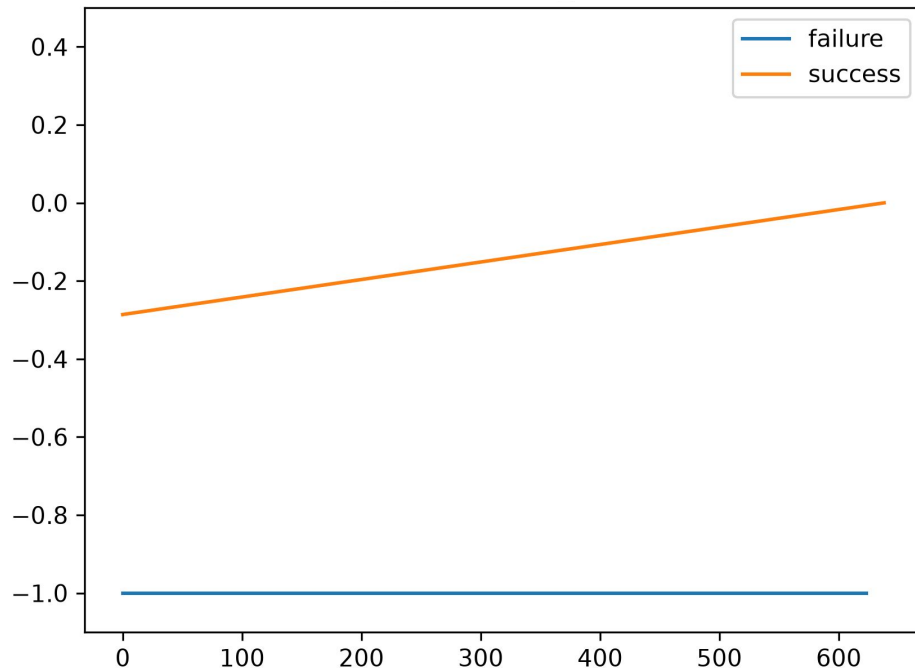
How to get ground truth values?

Given episode labeled as failure/success

$$r_t = \begin{cases} 0 & \text{if } t = T \text{ and success} \\ -C_{\text{fail}} & \text{if } t = T \text{ and failure} \\ -1 & \text{otherwise.} \end{cases}$$

+ clip value to (-1,0)

simplify: we don't care about when the robot will be unplugged after a failure



 **Datasets:**  DreamMachines/**20h_fullft_eval_success** 

Modalities:



Tabular



Time-series



Video

Formats:



parquet

Size:

10

800 unrolls (in the lightbox)

423 success (53%)

377 failures (47%)

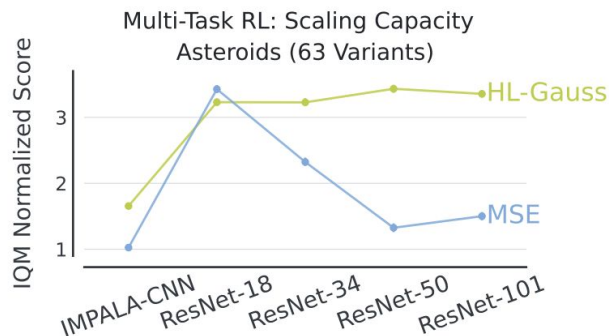
Stop Regressing: Training Value Functions via Classification for Scalable Deep RL

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^{*}Equal Contribution, [†]Core Contribution, ¹Google DeepMind, ²Mila, McGill University, ³Mila, Université de Montréal

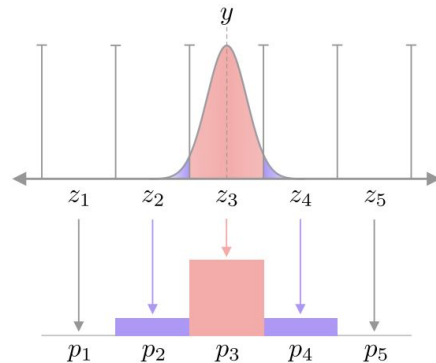
Value functions are a central component of deep reinforcement learning (RL). These functions, parameterized by neural networks, are trained using a mean squared error regression objective to match bootstrapped target values. However, scaling value-based RL methods that use regression to large networks, such as high-capacity Transformers, has proven challenging. This difficulty is in stark contrast to supervised learning: by leveraging a cross-entropy classification loss, supervised methods have scaled reliably to massive networks. Observing this discrepancy, in this paper, we investigate whether the scalability of deep RL can also be improved simply by using classification in place of regression for training value functions. We demonstrate that value functions trained with categorical cross-entropy significantly improves performance and scalability in a variety of domains. These include: single-task RL on Atari 2600 games with SoftMoEs, multi-task RL on Atari with large-scale ResNets, robotic manipulation with Q-transformers, playing Chess without search, and a language-agent Wordle task with high-capacity Transformers, achieving *state-of-the-art results* on these domains. Through careful analysis, we show that the benefits of categorical cross-entropy primarily stem from its ability to mitigate issues inherent to value-based RL, such as noisy targets and non-stationarity. Overall, we argue that a simple shift to training value functions with categorical cross-entropy can yield substantial improvements in the scalability of deep RL at little-to-no cost.

ResNet101

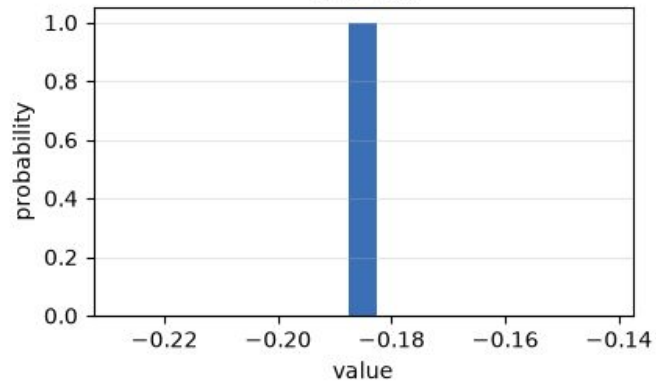


Smooth CE Loss

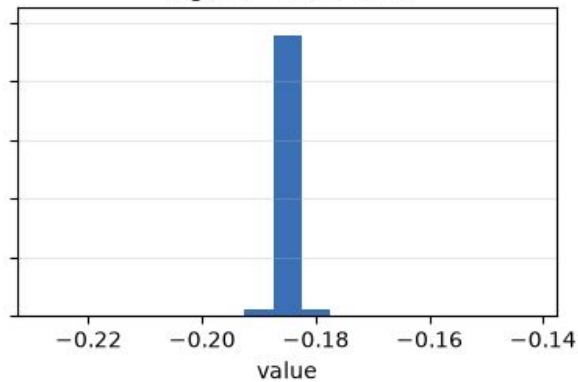
HL-Gauss



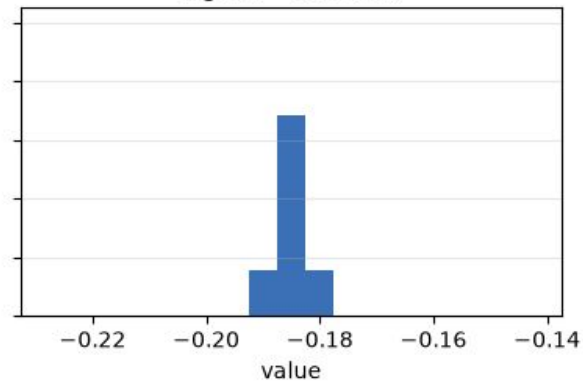
one-hot



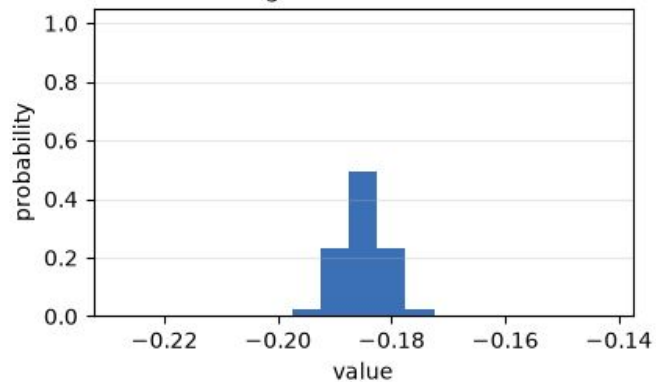
sigma = 0.25 bins



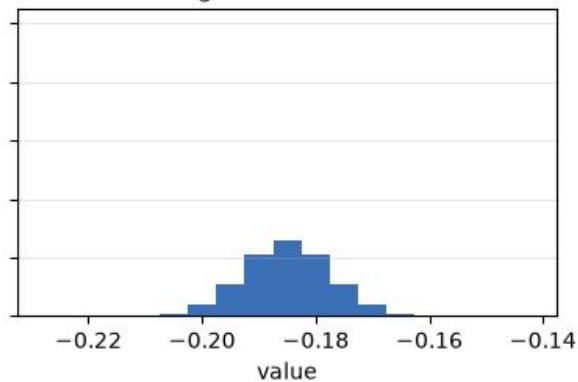
sigma = 0.5 bins



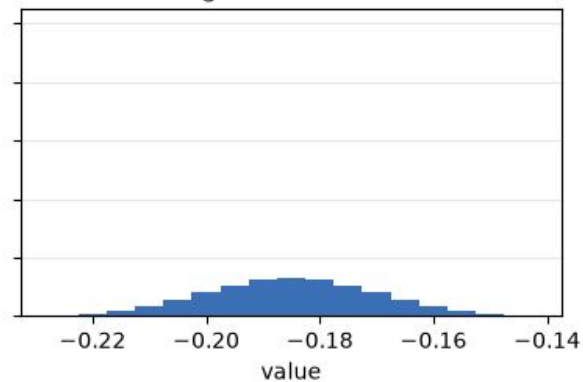
sigma = 0.75 bins



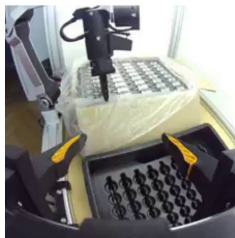
sigma = 1.5 bins



sigma = 3.0 bins



Train



ResNet
101

ResNet
101

ResNet
101

concat

2-layers
GeLU MLP

201 bins logits

Test

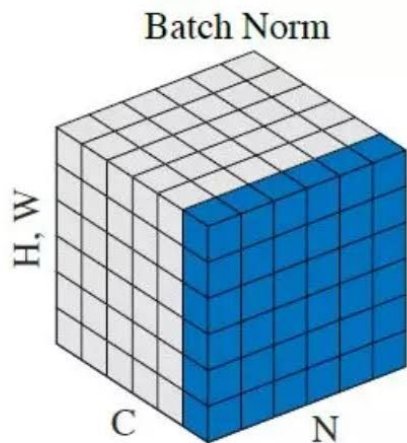
compute expected value via distributional value function

$$V^{\pi_{\text{ref}}}(o_t, \ell) \\ = \sum_{b \in [0, B]} p_{\phi}(V = b | \mathbf{o}_t) v(b)$$

train=600 eps

val=100 eps

test=100 eps



gradient accumulation with smaller batch sizes to parallelize overnight

ResNet-101 uses **BatchNorm** and computes statistics per forward pass...

left wrist



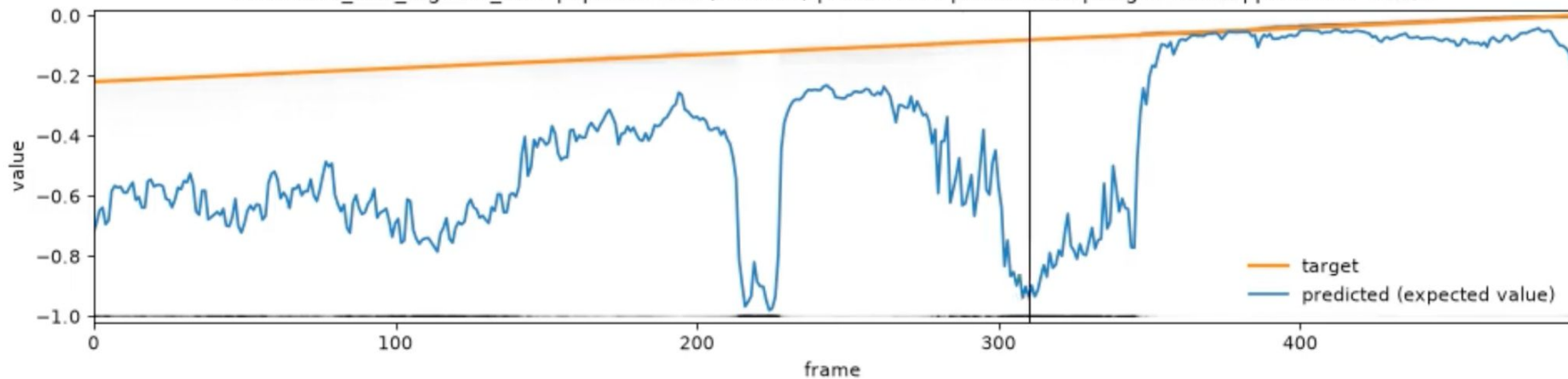
right wrist



top



resnet101_ffnn_hlgauss_0.75 | episode 434 (success) | MAE 0.303 | frame 310 | target -0.081 | predicted -0.940



left wrist



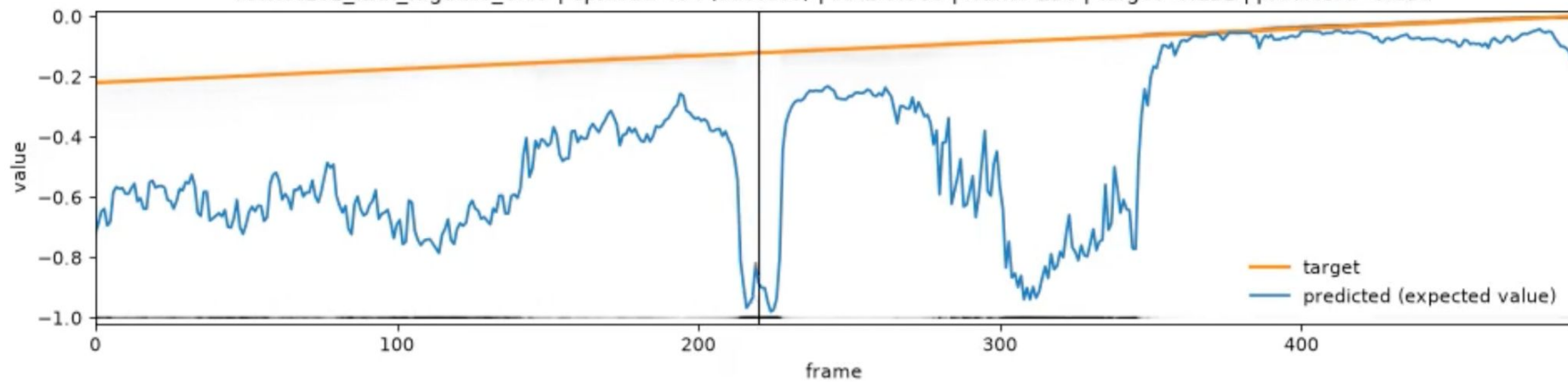
right wrist



top



resnet101_ffnn_hlgauss_0.75 | episode 434 (success) | MAE 0.303 | frame 220 | target -0.121 | predicted -0.885



left wrist



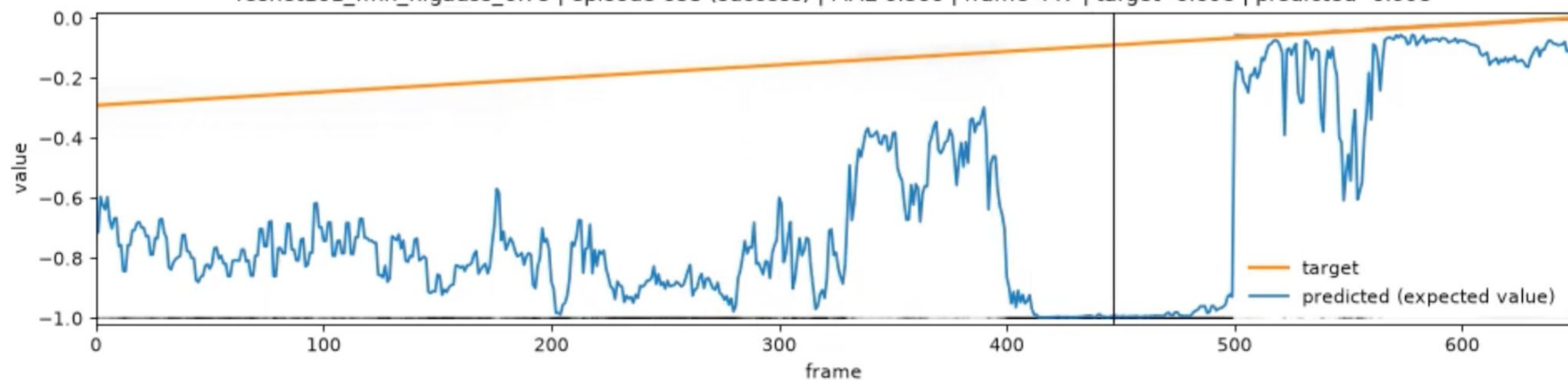
right wrist



top

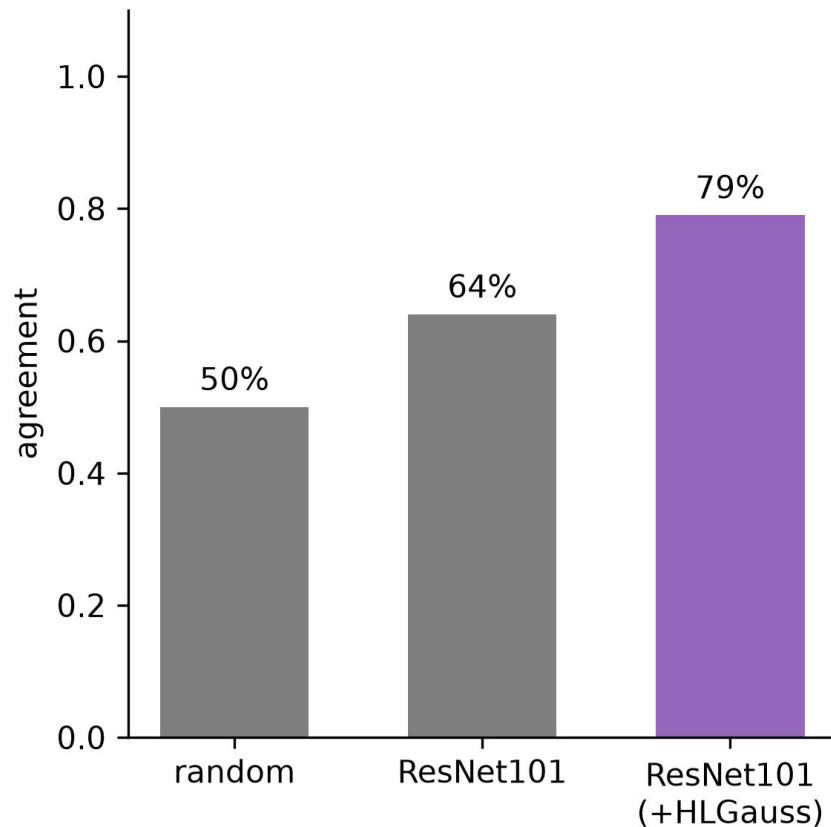


resnet101_ffnn_hlgauss_0.75 | episode 653 (success) | MAE 0.500 | frame 447 | target -0.090 | predicted -0.990



Quantitative Evaluation

1. N held-out episodes
2. Compute **value function**
3. Compute **advantage** for 1s clips
4. With 50% chance:
 - a. pick the **best** and **worst** clip by advantage
 - b. pick two **random** clips
5. **Rank** them **blindly**
6. Measure **agreement** between **human** and **advantage**-based ranking



* result for N=14

Next

1. Annotate more episodes for evaluation
2. Develop proxy metrics to grasp advantage accuracy without manual annotation during training
3. Train on more diverse data and tasks

Backup

SUCCESS episode 540, $t = 2 / 423$



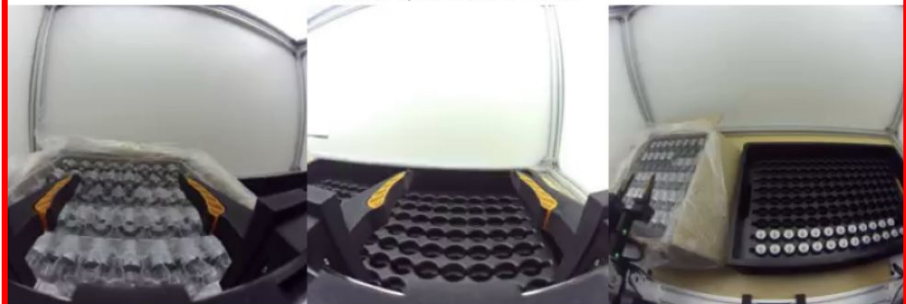
SUCCESS episode 1116, $t = 1 / 556$



SUCCESS episode 896, $t = 2 / 367$



FAILURE episode 1491, $t = 2 / 467$



FAILURE episode 1414, $t = 1 / 406$



FAILURE episode 1907, $t = 4 / 612$

